

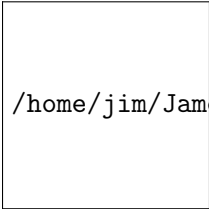
Section 3.3 part 2: "And"

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Monday 13th October, 2014



`/home/jim/JamesAshe/MHU/images/MarsHillU`

Section 3.3 part 2: "And"

Venn Diagrams and the Product Rule

Section 3.3 and 3.6

Dr. James Ashe

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Review/Preview: 3 “and” Problems

Example 3.3 part 2.1.

John has green, red, and blue pants; and a green and red shirt. What is the probability that he randomly selects blue pants **and** a blue shirt?

Review/Preview: 3 “and” Problems

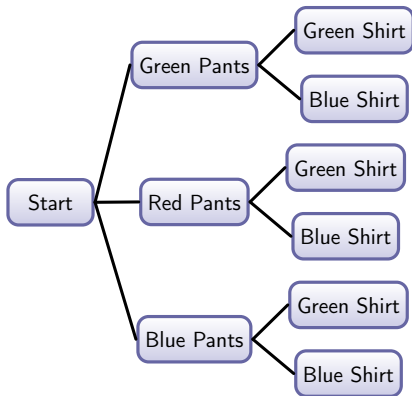
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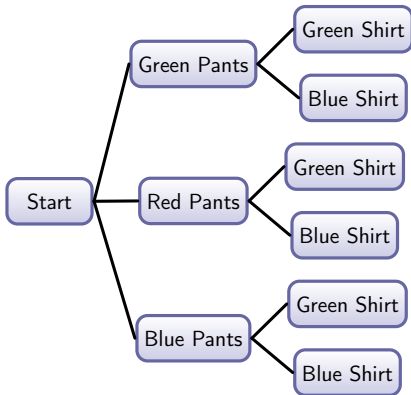
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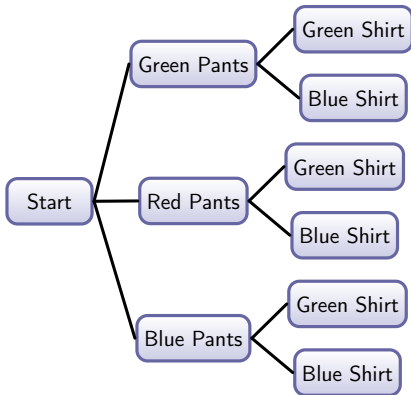


$$\begin{aligned} P(BP \cap BS) \\ = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6} \end{aligned}$$

Review/Preview: 3 "and" Problems

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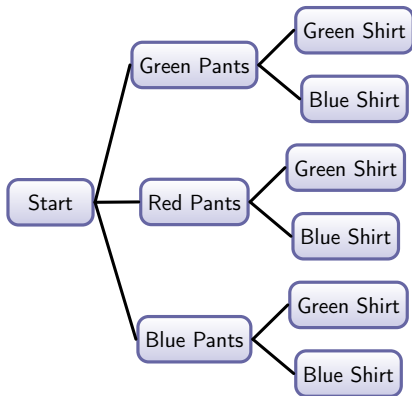
$$P(BP \cap BS) \\ = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

- The 1st event does not affect the probability of the 2nd event.

Review/Preview: 3 "and" Problems

Example 3.3 part 2.1.

John has green, red, and blue pants; and a green and red shirt. What is the probability that he randomly selects blue pants **and** a blue shirt?



$$P(BP \cap BS) \\ = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

- The 1st event does not affect the probability of the 2nd event.
- These events are called *independent*.

Review/Preview: 3 “and” Problems

Find the probability of the following:

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Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

Review/Preview: 3 “and” Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52}$$

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

Review/Preview: 3 “and” Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52} \times \frac{13}{52}$$

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

Example 3.3 part 2.4.

A king **and then** a heart is drawn without replacement.

Review/Preview: 3 “and” Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52} \times \frac{13}{52} = \frac{1}{52}$$

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A king **and then** a heart is drawn with replacement.

Example 3.3 part 2.4.

A king **and then** a heart is drawn without replacement.

Review/Preview: 3 "and" Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52} \times \frac{13}{52} = \frac{1}{52}$$

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

$$\frac{4}{52} \times \frac{13}{52}$$

Example 3.3 part 2.4.

A king **and then** a heart is drawn without replacement.

Conditional Probability.

$$p(A | B) = \frac{n(A \cap B)}{n(B)}$$

Probability of A if B has occurred.

Review/Preview: 3 "and" Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52} \times \frac{13}{52} = \frac{1}{52}$$

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

$$\frac{4}{52} \times \frac{13}{52} = \frac{52}{2704} = \frac{1}{52}$$

Example 3.3 part 2.4.

A king **and then** a heart is drawn without replacement.

Conditional Probability.

$$p(A | B) = \frac{n(A \cap B)}{n(B)}$$

Probability of A if B has occurred.

Product Rule

$$p(A \cap B) = p(A | B) \cdot p(B)$$

Review/Preview: 3 "and" Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52} \times \frac{13}{52} = \frac{1}{52}$$

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

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Example 3.3 part 2.4.

A king **and then** a heart is drawn without replacement.

drawing a non-heart king and heart

OR

drawing a king of hearts and a heart.

Conditional Probability.

$$p(A | B) = \frac{n(A \cap B)}{n(B)}$$

Probability of A if B has occurred.

Product Rule

$$p(A \cap B) = p(A | B) \cdot p(B)$$

Review/Preview: 3 "and" Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

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Example 3.3 part 2.4.

A king **and then** a heart is drawn without replacement.

drawing a non-heart king and heart

OR

drawing a king of hearts and a heart.

$$\left(\frac{3}{52} \times \frac{13}{51}\right) + \left(\frac{1}{52} \times \frac{12}{51}\right)$$

Conditional Probability.

$$p(A | B) = \frac{n(A \cap B)}{n(B)}$$

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Product Rule

$$p(A \cap B) = p(A | B) \cdot p(B)$$

Review/Preview: 3 "and" Problems

Find the probability of the following:

Example 3.3 part 2.2.

A king **and** a heart is drawn.

$$\frac{4}{52} \times \frac{13}{52} = \frac{1}{52}$$

Example 3.3 part 2.3.

A king **and then** a heart is drawn with replacement.

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drawing a non-heart king and heart

OR

drawing a king of hearts and a heart.

$$\left(\frac{3}{52} \times \frac{13}{51}\right) + \left(\frac{1}{52} \times \frac{12}{51}\right) = \frac{51}{2652} \approx 0.019$$

Conditional Probability.

$$p(A | B) = \frac{n(A \cap B)}{n(B)}$$

Probability of A if B has occurred.

Product Rule

$$p(A \cap B) = p(A | B) \cdot p(B)$$

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Find the probability of the following:

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Review/Preview: 3 “and” Problems

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Example 3.3 part 2.5.

A king **and** heart is drawn.

Review/Preview: 3 “and” Problems

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A king **and** heart is drawn.

Review/Preview: 3 “and” Problems

Find the probability of the following:

Example 3.3 part 2.5.

A king **and** heart is drawn.

Example 3.3 part 2.6.

A king **and then** a heart is drawn with replacement.

Review/Preview: 3 “and” Problems

Find the probability of the following:

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Review/Preview: 3 “and” Problems

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Example 3.3 part 2.5.

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A king **and then** a heart is drawn with replacement.

Example 3.3 part 2.7.

A king **and then** a heart is drawn without replacement.

Review/Preview: 3 “and” Problems

Find the probability of the following:

Example 3.3 part 2.5.

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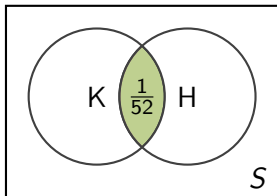
Review/Preview: 3 “and” Problems

Find the probability of the following:

Example 3.3 part 2.5.

A king **and** heart is drawn.

$$P(K \cap H) = \frac{1}{52}$$



$$P(K \cap H)$$

Example 3.3 part 2.6.

A king **and then** a heart is drawn with replacement.

Example 3.3 part 2.7.

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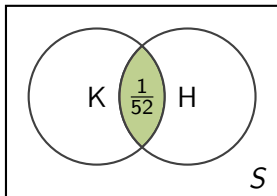
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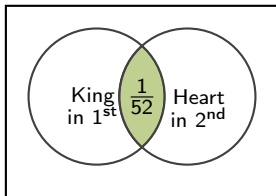


$$P(K \cap H)$$

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A king **and then** a heart is drawn with replacement.

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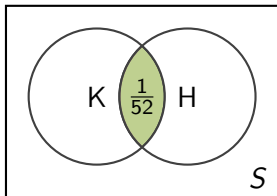
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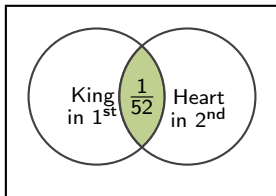


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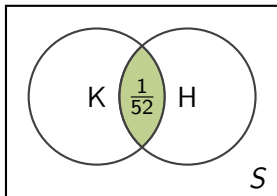
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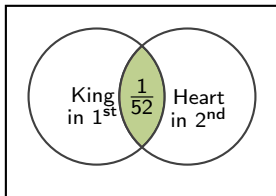


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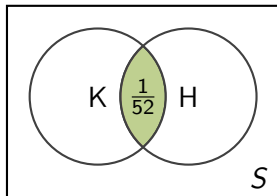
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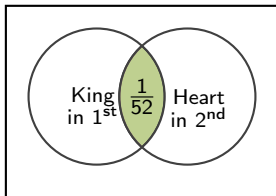


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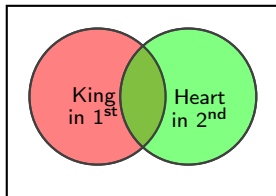


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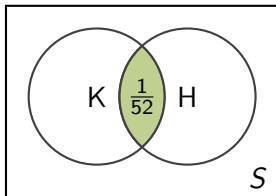
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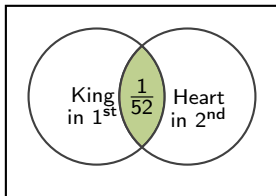


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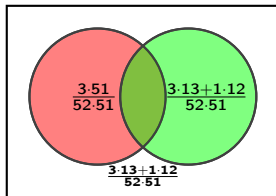


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$$P(K \cap H)$$

Probabilities and Venn Diagrams

Probabilities and Venn Diagrams

Probability that a student does NOT fail Math: $1 - .65$

Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

Probability that a student does NOT fail Math: $1 - .65 = .35$

Probabilities and Venn Diagrams

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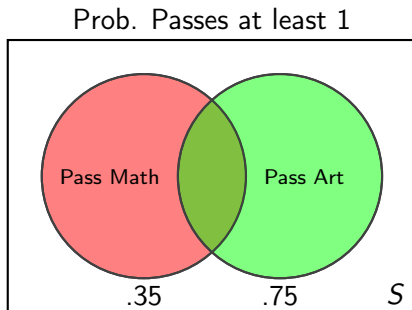
- a. passes Math.

Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.



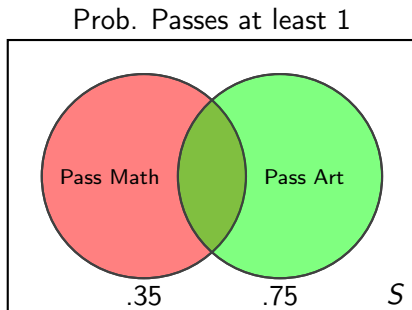
Probability that a student passes Math AND Art is: $P(M \cap A)$

Probabilities and Venn Diagrams

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Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both ↓

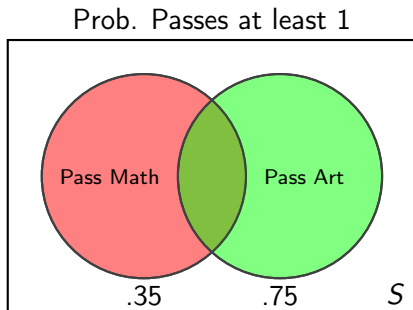
$$\begin{aligned} P(M \cup A) &= P(M) + P(A) - P(M \cap A) \\ .85 &= \end{aligned}$$

Probabilities and Venn Diagrams

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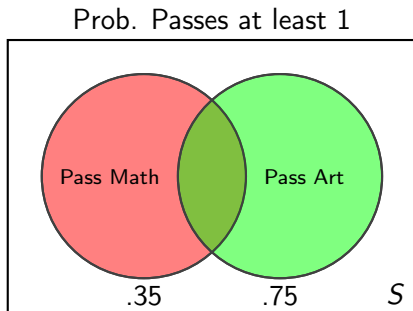
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Probabilities and Venn Diagrams

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Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both ↓

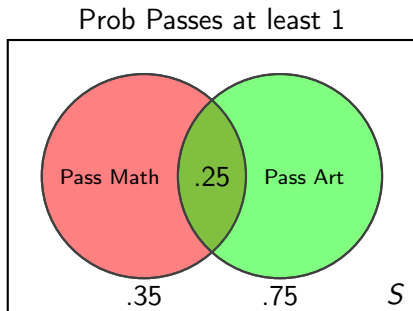
$$\begin{array}{rclclcl} P(M \cup A) & = & P(M) & + & P(A) & - & P(M \cap A) \\ .85 & = & .35 & + & .75 & - & \end{array}$$

Probabilities and Venn Diagrams

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The probability that a student passes Art is 75%, fails Math is 65%, and passes at least one is 85%. Find that a student:

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Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both ↓

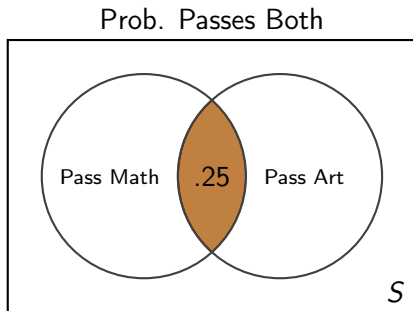
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Probability that a student passes Math AND Art is: $P(M \cap A)$

Pass both↓

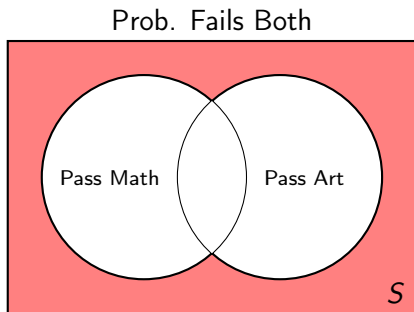
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- a. passes Math.
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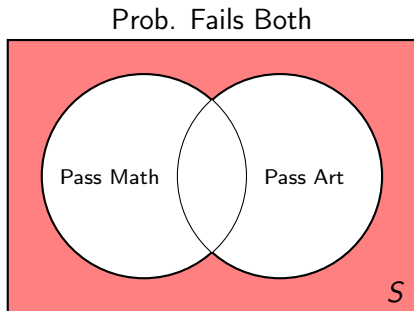


Probabilities and Venn Diagrams

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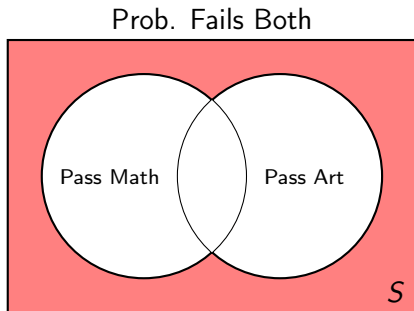
Probability that a student does NOT pass at least one.

Probabilities and Venn Diagrams

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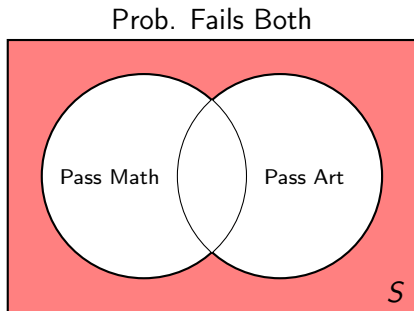
Probability that a student does NOT pass at least one.
 $1 - .85$

Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.



Probability that a student does NOT pass at least one.

$$1 - .85 = .15$$

Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.
- d. passes only one.

Probabilities and Venn Diagrams

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The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

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- d. passes only one.

Probability that a student passes just Math

Probabilities and Venn Diagrams

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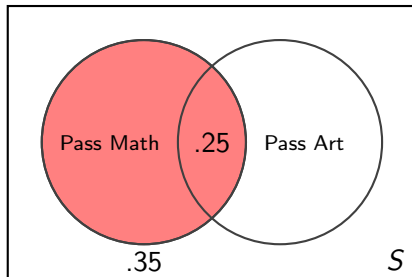
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- b. passes both.
- c. fails both.
- d. passes only one.

Probability that a student passes just Math

$$P(M) - P(M \cap A)$$

Prob. Passes Math = .35



Probabilities and Venn Diagrams

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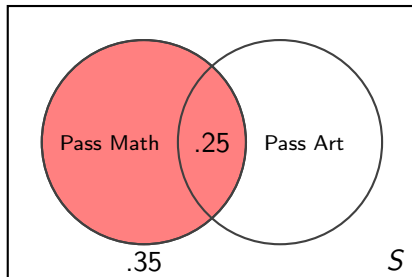
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Probability that a student passes just Math

$$P(M) - P(M \cap A)$$

$$.35 - .25$$

Prob. Passes Math = .35



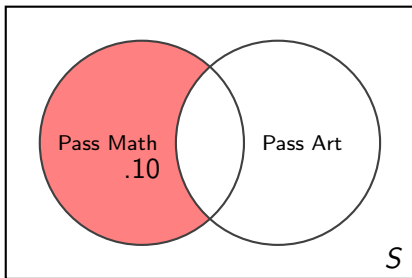
Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student passes Art is 75%, fails Math is 65%, and passes at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.
- d. passes only one.

Prob. Passes Only Math



Probability that a student passes just Math

$$\begin{array}{rclcl} P(M) & - & P(M \cap A) & & \\ .35 & - & .25 & = & .10 \end{array}$$

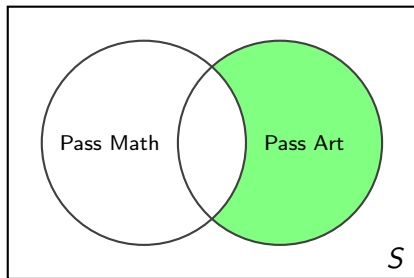
Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.
- d. passes only one.

Prob. Passes Only Art



Probability that a student passes just Math OR just Art

$$P(A) - P(M \cap A)$$

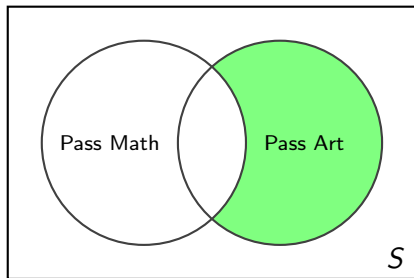
Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student passes Art is 75%, fails Math is 65%, and passes at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.
- d. passes only one.

Prob. Passes Only Art



Probability that a student passes just Math OR just Art

$$\begin{aligned} P(A) &= P(M \cap A) \\ .75 &= .25 = .5 \end{aligned}$$

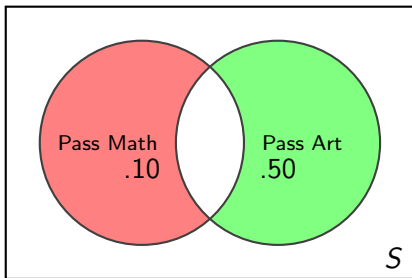
Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student passes Art is 75%, fails Math is 65%, and passes at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.
- d. passes only one.

Prob. Passes only Math or Art



Probability that a student passes just Math OR just Art

$$\begin{array}{rcl} P(A) & - & P(M \cap A) \\ .75 & - & .25 \\ & & = .5 \end{array}$$

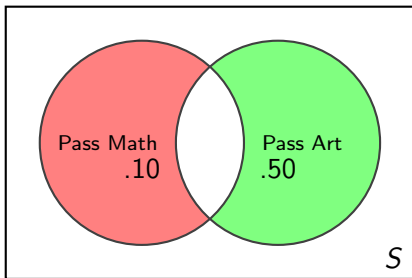
Probabilities and Venn Diagrams

Example 3.3 part 2.8.

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- b. passes both.
- c. fails both.
- d. passes only one.

Prob. Passes only Math or Art



Probability that a student passes just Math OR just Art
 $.10 + .50$

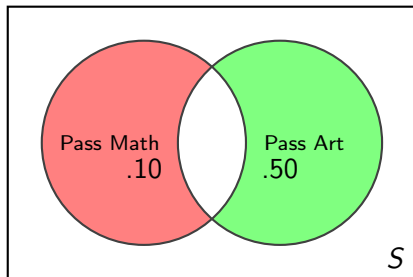
Probabilities and Venn Diagrams

Example 3.3 part 2.8.

The probability that a student *passes* Art is 75%, *fails* Math is 65%, and *passes* at least one is 85%. Find that a student:

- a. passes Math.
- b. passes both.
- c. fails both.
- d. passes only one.

Prob. Passes only Math or Art



Probability that a student passes just Math OR just Art
 $.10 + .50 = .60$